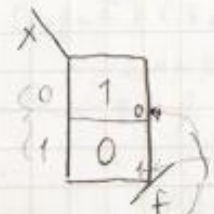
KARNAUGH MAPS (K-Map)

1). first order K-Map

Dec	X	$F = \bar{X}$
0	0	1
1	1	0



$f_{sop} = \bar{x}$ $\quad f_{sop} = \sum m(0)$

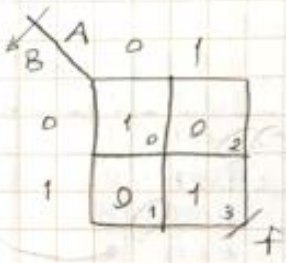
$f_{pos} = \bar{x}$ $\quad f_{pos} = \prod M(1)$

Algebra $\quad$ Numerik

2) second order K-Map

Dec	A	B	f
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1





$$f_{SOP} = \bar{A}\bar{B} + AB$$

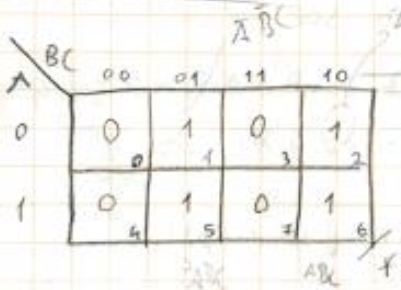
$$f_{POS} = (\bar{A}+B)(A+\bar{B})$$

$$f_{SOP} = \Sigma m(0,3)$$

$$f_{POS} = \Pi M(1,2)$$

Third-order K-Map

A	B	C	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



=> GRAY CODE



$$f_{SOP} = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$$

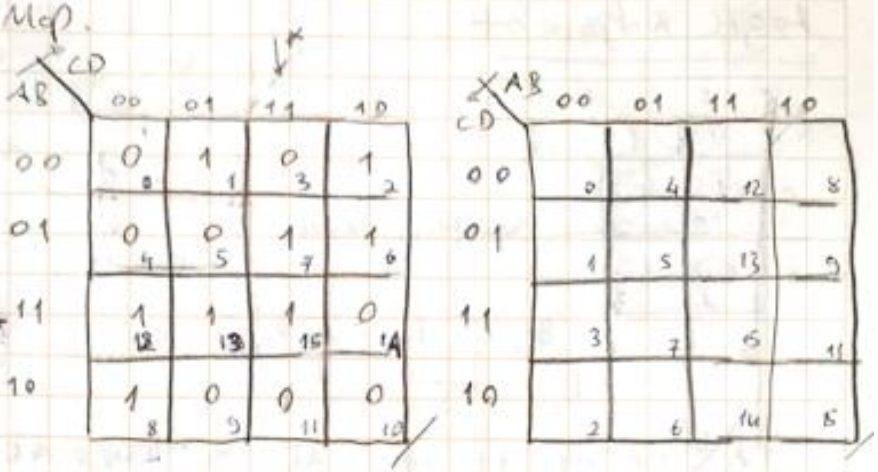
$$f_{POS} = (A+B+C)\cdot(A+\bar{B}+\bar{C})\cdot(\bar{A}+B+C)\cdot(\bar{A}+\bar{B}+\bar{C})$$

$$f_{SOP} = \Sigma m(1,2,5,6)$$

$$f_{POS} = \Pi M(0,3,4,7)$$

fourth - order K-Map.

Dez	A B C D	f
0	0 0 0 0	0
1	0 0 0 1	1
2	0 0 1 0	1
3	0 0 1 1	0
4	0 1 0 0	0
5	0 1 0 1	0
6	0 1 1 0	1
7	0 1 1 1	1
8	1 0 0 0	1
9	1 0 0 1	0
10	1 0 1 0	0
11	1 0 1 1	0
12	1 1 0 0	1
13	1 1 0 1	1
14	1 1 1 0	0
15	1 1 1 1	1

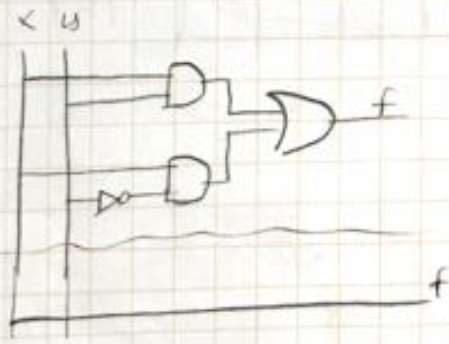


K-MAP FUNCTION REDUCTION = MINIMIZATION

Logic Adjacencies / logik komşuluk

$A\bar{B}C + \bar{A}\bar{B}C$ } 2 AND, 1 OR, 2 inverter / 11 input $\quad xy + x\bar{y} = x.(y+\bar{y}) = x.1 = x$

$Ax + \bar{A}x = (A+\bar{A})x = 1.x = x = \bar{B}C$ → 1 AND + 1 inverter / 3 input



2 AND
1 OR
1 inverter
+7 input (Line) = Gates / inputs

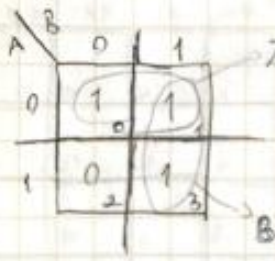
→ 1

Cost criteria

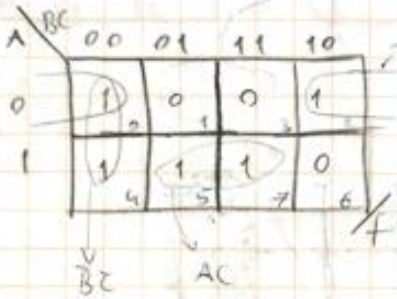
Gates / inputs.

OSOW

Logic Adjacent



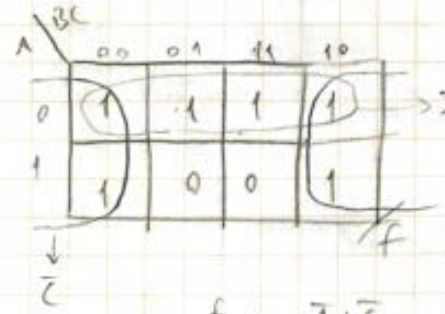
$f_{sop} = \bar{A} + B$



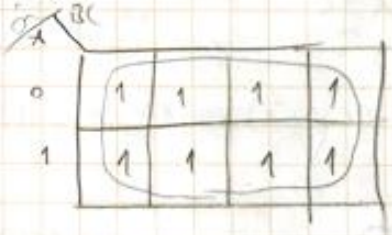
$f_{sop} = \bar{A}\bar{C} + \bar{B}\bar{C} + AC$

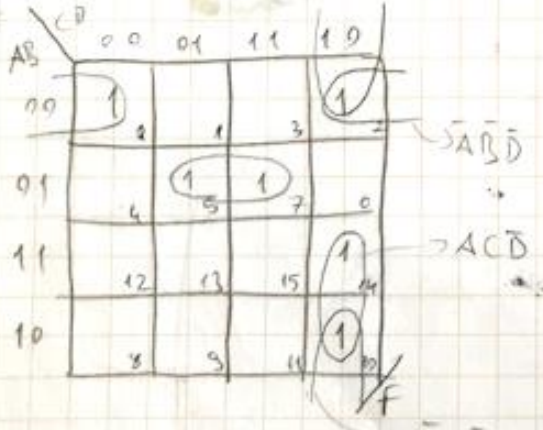
2 seçilde kalıyorsa ve hepsini alıyorsa

$f_{pos} = (A + \bar{C}) \cdot (\bar{A} + \bar{B} + C)$

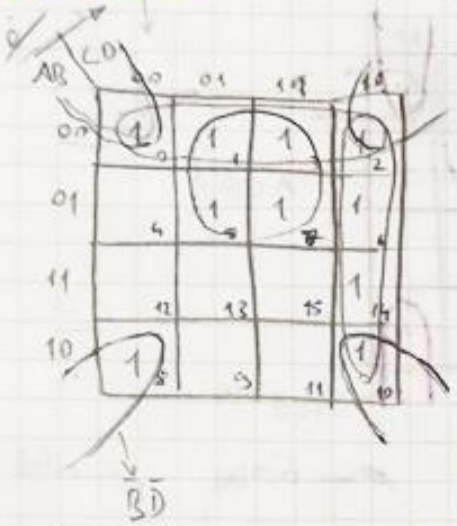


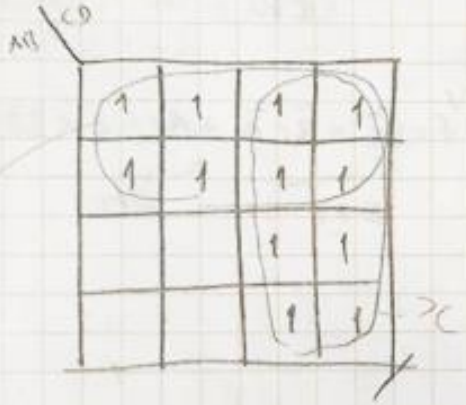
$f_{sop} = \bar{A} + \bar{C}$

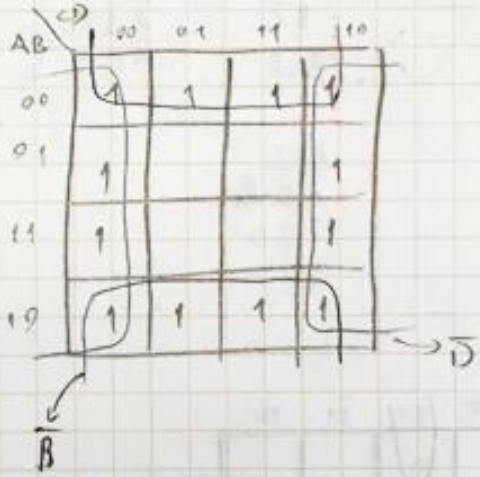


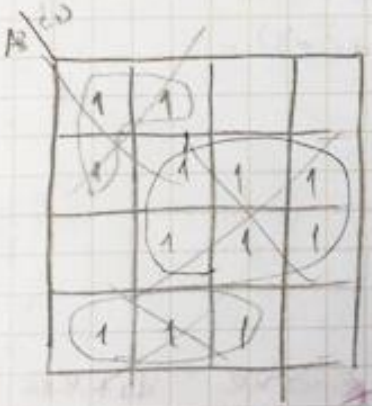


OSOW









→ (redundant)

Prime Implicant (PI)

Essential Prime Implicants (EPI)

Optional Prime Implicants (OPI)

Redundant " " (RPI)

(überflüssig)